

Question	Marking Guidance	Mark	Comments
04.1	1. Reduces activation energy; 2. Due to bending bonds OR Without enzyme, very few substrates have sufficient energy for reaction;	2	1. Accept 'reduces E_a'. 2. Accept 'Due to stress/pressure/tension on bonds' OR 'Due to weakening bonds'. 2. Ignore references to 'breaking bonds'.
04.2	1.93×10^{11};; Allow 1 max for $578/3.0 \times 10^{-9}$ 1.93×10^x when $x \neq 11$ Correct answer with incorrect standard form e.g. 19.3×10^{10}	2	Accept any number of significant figures ≥ 2, if rounding correct (1.926×10^{11}). Same principle applies to one max answers.
04.3	31.4;; Allow 1 max for 0.44 and 1.4 32.8 33.1 30 29.3	2	Accept any number of significant figures ≥ 2, if rounding correct (31.4284714). Same principle applies to 1 max answers. 32.8 = Both readings at 2.5 mmol dm^{-3} (0.44/1.34) 33.1 = Both readings at 2.5 mmol dm^{-3} (0.44/1.33) 30 = Incorrect reading for C (0.42/1.4) 29.3 = Incorrect reading for C (0.41/1.4)

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04.4	1. (Binding) alters the tertiary structure of the enzyme ; 2. (This causes) <u>active site</u> to change (shape); 3. (So) More (successful) E-S complexes form (per minute) OR E-S complexes form more quickly OR Further lowers activation energy;	3	Max 1 if lyxose acting as an inhibitor OR if answer linked to lower rate of reaction OR if lyxose used an energy source/respiratory substrate 3. Accept 'acts as a co-enzyme' 3. Accept description for E-S complexes.

